Networking Fundamentals

Understanding IP addresses, MAC addresses, subnetting, security protocols, and network protection

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What is an IP Address?

IP address stands for internet protocol. It's a unique number that identifies network-connected devices.

192.168.0.2 IP address dividing 4 Octect or dotted representing.

Two parts of IP address: Network part and Host part.

Example: Like your home wifi 192.168.0.1, laptop 192.168.0.2, printer 192.168.0.3, mobile.

This IP address is considered first 3 parts is network part and last part host part or (how many devices connected that network).

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Types of IP Address

Public IP

This is the address assigned to your network router by your Internet Service Provider (ISP). It is unique across the entire internet and is how the outside world identifies your home or business.

Private IP

This is the address your router assigns to your specific devices (phone, laptop, smart fridge) within your local network. These aren't visible to the internet; they stay within your local network.

Dynamic vs Static

Dynamic

Changed automatically by the network (via DHCP). Most common for home users. Standard browsing, smartphones.

Static

Manually assigned and never changes. Usually costs extra from an ISP. Hosting servers, gaming, or VoIP services.

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IPv4 vs IPv6

Feature	IPv4	IPv6
Address Size	32-bit	128-bit
Address Format	Numeric (Dotted Decimal)	Alphanumeric (Hexadecimal)
Number of Addresses	~4.3 Billion (2^{32})	~340 Undecillion (2^{128})
Configuration	Manual or via DHCP	Self-configuration (SLAAC) or DHCPv6
Security	Optional (IPsec)	Built-in (IPsec support required)
Header Size	20 to 60 bytes (Variable)	40 bytes (Fixed)

01

Address Capacity

The biggest driver for IPv6 is scarcity. IPv4 uses 32-bit addresses, which looks like 192.168.1.1. Because the internet exploded in popularity, we've effectively run out of these addresses. IPv6 uses 128-bit addresses, formatted like 2001:0db8:85a3:0000:0000:8a2e:0370:7334.

02

Header Efficiency

IPv6 headers are designed to be "cleaner." While the address is longer, the header structure is fixed. This allows routers to process packets faster because they don't have to check for optional fields as they do with IPv4.

03

No More NAT

In IPv4, your home router usually has one public IP, and it "hides" all your devices behind it using NAT. IPv6 provides so many addresses that every single device in your house can have its own unique, public IP address, simplifying peer-to-peer connections and VoIP.

04

Improved Security and Mobility

IPv6 was built with the modern mobile world in mind. It has better support for "Mobile IP" (keeping a connection while moving between Wi-Fi and Cellular) and was designed from the ground up to support IPsec, which encrypts and authenticates your data.

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What is a MAC Address

Think of a MAC Address (Media Access Control) as the "Social Security Number" for your hardware. While an IP address changes depending on which Wi-Fi network you join, the MAC address is "burned" into your network interface card (NIC) at the factory and stays with the device forever.

Anatomy of a MAC Address

A MAC address is a 48-bit identifier usually written as six groups of two hexadecimal digits (e.g., 00:1A:2B:3C:4D:5E). It is divided into two parts:

- **OUI (Organizationally Unique Identifier):** The first three groups (24 bits) identify the manufacturer (e.g., Apple, Samsung, Intel).
- **NIC Specific:** The last three groups (24 bits) are a unique serial number assigned by that manufacturer.

Types of MAC Addresses

Not all MAC addresses are meant for a single device. They are categorized based on how they communicate with the network:

Unicast MAC Address

This is the most common type. It identifies a single specific device on the network. When a frame is sent to a Unicast address, it is intended for only one pair of eyes. **Example:** Sending a file from your laptop to your printer.

Multicast MAC Address

This address represents a group of devices or services. Devices can "subscribe" to a multicast group to receive specific data. **How to spot it:** In Ethernet, these always begin with the prefix 01:00:5E. **Example:** A video stream being sent to multiple employees in an office simultaneously.

Broadcast MAC Address

This is the "shout-out" address. When a frame is sent here, every device on the local network segment receives and processes it. **The Format:** It is always all "Fs": FF:FF:FF:FF:FF:FF. **Example:** An ARP Request (which we discussed earlier) uses a broadcast address to find who owns a specific IP.

How to Find Your MAC Address

If you're curious about your own hardware's "fingerprint," you can find it quickly:

- **Windows:** Open Command Prompt and type getmac or ipconfig /all.
- **macOS/Linux:** Open Terminal and type ifconfig (look for "ether").
- **Android/iOS:** Look under Settings > About Phone > Status or General >

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Difference Between MAC Address and IP Address

Feature	MAC Address (Media Access Control)	IP Address (Internet Protocol)
Full Name	Physical Address / Hardware Address	Logical Address / Network Address
Uniqueness	Hard-coded into the device at the factory	Assigned by the network or ISP
Changeability	Permanent (mostly)	Temporary (changes with networks)
Layer (OSI)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Scope	Used for local communication (LAN)	Used for global communication (Internet)
Format	Hexadecimal (e.g., 00:1A:2B:3C:4D:5E)	Numeric (e.g., 192.168.1.1)

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ARP Protocol

ARP protocol stands for address resolution protocol. It is used to map an IP address to MAC address within a local network.

Step 1: ARP Request

Broadcast who is IP address.

Step 2: ARP Reply

Device with that IP address sends its MAC address.

ARP Cache: IP-MAC mapping is stored in the ARP Cache.

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Subnetting

Subnetting is the process of dividing larger network into smaller network or subnet.

Improve Network Performance

Enhanced Security

Like firewall or access control list. This working is different department using different subnet placed. Example: guest wifi subnet and HR server subnet. A guest subnet only access for only providing subnet. HR server blocked that guest subnet blocked.

Troubleshoot

Preventing Waste of IP Address

Subnetting Providing Different Classes

Class	Range of First Octet	Purpose	Default Subnet Mask
Class A	1 – 127	Huge networks (governments, ISPs)	255.0.0.0
Class B	128 – 191	Medium networks (large universities/corps)	255.255.0.0
Class C	192 – 223	Small networks (home/small business)	255.255.255.0
Class D	224 – 239	Multicasting (not for hosts)	N/A
Class E	240 – 255	Experimental/Research	N/A

This class is using identify network part and host part.

```
Class A: 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0
Class B: 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0
Class C: 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
```

The Seven Characteristics for Subnetting

- Network ID
  This is the first IP address in each subnet
- Broadcast ID
  This is the last IP address in each subnet
- First Host ID
  Represents the IP address after the Network ID
- Last Host ID
  Representing the IP address before broadcast ID
- Next Network
  This define the network ID of the next sub-network
- #IP Address
  The total no. of IP addresses in the sub-network

Example: 192.168.0.0/25

Subnetting

- Subnet: 255.255.255.128
- Net ID: 192.168.0.0
- Broadcast ID: 192.168.0.127
- First Host IP: 192.168.0.1
- Last Host IP: 192.168.0.127
- Next IP address: 192.168.0.128

CIDR Notation

CIDR (Classless Inter-Domain Routing) is the standard system used to define IP addresses and their associated network masks. Instead of using complex "dotted-decimal" subnet masks (like 255.255.255.0), CIDR uses a simple slash notation to indicate how many bits are fixed for the network portion of the address.

How to Find Host Range, Network Bit, Host Bit, Subnet Mask

Example: 192.168.50.0/26

- Network bit = 26
- Host bit = 6

```
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0
128 64 32 24 16 8 4 2 1
```

- Network mask = 255.255.255.192 (128+64)
- Host range 2 * 2 * 2 * 2 * 2 * 2 = 64 - 2 = 62
- Network ID = 192.168.50.0
- 192.168.50.1 Usable host to 192.168.50.62
- Broadcast ID = 192.168.50.63

Department Example

HR = 60, IT = 28, Med = 35

HR Department = 60+2

- 192.168.50.0
- 192.168.50.1 Usable IP
- 192.168.50.62
- 192.168.50.63

Medical Department = 35

- 192.168.50.64
- 192.168.50.65 Usable IP
- 192.168.50.126
- 192.168.50.127

IT Department

- 192.168.50.128
- 192.168.50.129
- 192.168.50.158
- 192.168.50.159

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SSL and TLS

SSL

SSL stands for Secure Socket Layer. It is a communication protocol or set of rules that create a secure connection between two devices or applications on a network. SSL is the process security flaws.

TLS

TLS stands for Transport Layer Security. It is an upgraded version that fixes existing SSL vulnerabilities. TLS authentication is more efficient and continues to support encrypted communication channels.

TLS Handshaking

Client and Server Working TLS Handshaking

Client sends hello. That list of information such as the SSL and TLS version, cryptographic algorithm, data compression method supported by the client.

Server Response Reply Hello

That contains cryptographic algorithm chosen, session ID, sends its digital certificate and publicly.

Client Will Contact the Server CA

And verify the server digital certificate and authentication.

Client Key Exchange

The client sends a shared secret key to be used in the following conversation. The secret key is encrypted with the server public key.

Client Sends a Finished Message

Is encrypted to shared secret key handshake complete.

Server Response the Encrypted Stored Secret Key

Handshake complete.

The server and client can now exchange messages that are symmetrically encrypted with shared secret key.

IP Sec

IP sec is a set of protocols used to security at the network layer of the OSI model.

Two Types of Protocol

AH Authentication Header

IP sec protocol that offers authentication and integrity.

ESP Encapsulating Security Payload

This protocol offers authentication, integrity, and encryption.

Two Types of Mode

Transport Mode

Only the payload (the data portion) of the IP packet is encrypted/authenticated. The original IP header remains visible. **Best for:** End-to-end communication between two specific hosts (e.g., a client connecting to a server).

Tunnel Mode

The entire original IP packet (header + payload) is encrypted and encapsulated within a new, outer IP header. **Best for:** Site-to-site VPNs (gateway-to-gateway) or connecting a host to a gateway, as it hides the internal network topology from the public internet.

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IDS, IPS, VPN, and Proxy

IDS and IPS

IDS stands for Intrusion Detection System. Device or software application that monitors a network or system for malicious activity or policy violations. Any intrusion activity or violation typically reported to an administrator or collected centrally.

Two Types of IDS

Network Intrusion Detection System (NIDS)

A NIDS is placed at strategic points within the network—typically at the perimeter (near the firewall) or at the boundaries between network segments.

How it works: It analyzes traffic passing through the network segment it is monitoring. It looks at the packet headers and payloads of traffic flowing between devices.

Best for: Detecting broad network-wide attacks, such as DoS (Denial of Service) attacks, port scanning, and malware spreading across the network.

Limitation: Because it sits on the network wire, it cannot inspect traffic that is encrypted unless it has the decryption keys, and it cannot see what is happening inside a specific server or workstation.

Host-based Intrusion Detection System (HIDS)

A HIDS is software installed directly on an individual device, such as a server, workstation, or database host.

How it works: It monitors the internal activity of that specific host. This includes tracking system call logs, file integrity (checking if critical system files have been modified), application logs, and login attempts.

Best for: Detecting attacks that have already bypassed network defenses or internal threats from a user/process already logged into the system. It is excellent for identifying unauthorized changes to sensitive files.

Limitation: It only protects the device it is installed on. If you have 500 servers, you need to manage 500 individual agents, which can be resource-intensive.

An IPS is an extension of IDS. Because it is placed inline—meaning all network traffic must pass through it—it has the authority to make real-time decisions. If it identifies a threat based on its rule set, it can immediately drop the packets, block the source IP, or reset the connection to prevent the attack from succeeding.

Types of IPS

1

Network-based IPS (NIPS)

What it does: Monitors the entire network by analyzing traffic at strategic points, usually near the perimeter (behind the firewall) or between critical network segments.

Key Focus: It inspects packets and protocol activity to identify threats like DoS attacks, port scans, and network-wide malware propagation.

Best for: Broad, enterprise-level protection where you need to catch malicious traffic before it reaches your internal assets.

2

Host-based IPS (HIPS)

What it does: Installed directly on an individual endpoint (a server, workstation, or database).

Key Focus: It monitors activity specifically on that device—such as system calls, file integrity, and I/O operations—and blocks malicious traffic destined for that machine.

Best for: Protecting high-value targets (like a database server) or providing a "last line of defense" if a threat manages to bypass the NIPS.

3

Wireless IPS (WIPS)

What it does: Monitors your wireless network (Wi-Fi) specifically.

Key Focus: It acts as a gatekeeper for the radio spectrum, scanning for unauthorized access points (rogue APs), man-in-the-middle attacks, and brute-force attempts against wireless protocols.

Best for: Ensuring that the airwaves around your building remain secure and free from unauthorized entry points.

4

Network Behavior Analysis (NBA)

What it does: Rather than just looking for specific "bad" packets, it looks for unusual traffic patterns or flows across the network.

Key Focus: It is highly effective at identifying threats that don't rely on known signatures, such as a compromised machine suddenly starting a massive data exfiltration or a distributed denial-of-service (DDoS) attack.

Best for: Detecting sophisticated attacks, insider threats, and anomalies that traditional signature-based systems might miss.

VPN / Proxy

VPN

VPN stands for Virtual Private Network. It's a secure and encrypted connection between device and network.

Key Benefits

- **Enhanced Security:** Encryption protects your data from being intercepted, which is vital when using public Wi-Fi.
- **Privacy/Anonymity:** Your ISP cannot see which websites you visit; they only see that you are connected to a VPN server.
- **Geo-Unblocking:** By masking your location, you can access content or services that may be restricted in your current region.

Common Use Cases

- **Remote Work:** Providing employees secure access to internal company resources while working from home.
- **Public Wi-Fi Protection:** Securing browsing sessions in coffee shops, airports, or hotels where network traffic is easily sniffed.
- **Bypassing Censorship:** Accessing information in regions where certain websites or social media platforms are restricted.

The Privacy Reality: What a VPN Does and Doesn't Do

It is a common misconception that a VPN makes you "invisible." It is important to understand the limits:

What it protects: It hides your IP address and encrypts your traffic from your ISP and local network snoopers.

What it doesn't protect: If you log into a website (like Google or Facebook), that website still knows exactly who you are based on your credentials. Furthermore, you are shifting your trust from your ISP to the VPN provider—they can see all the traffic you send through their servers.

Disadvantages

- **Performance Latency:** Because your data must travel to a remote VPN server before reaching its final destination, you may experience slower internet speeds.
- **The "Trust" Factor:** You are essentially giving the VPN company the ability to monitor your traffic. Choosing a reputable provider with a verified "no-logs" policy is critical.
- **Blocked Connections:** Some websites and services actively block IP addresses associated with known VPN providers.

Proxy

A proxy server acts as an intermediary between your device and the internet. When you use one, your web requests don't go directly to the website; they go to the proxy, which then forwards the request on your behalf, receives the response, and sends it back to you.

Key Benefits

- **Anonymity:** The website sees the proxy's IP address, not yours.
- **Bypassing Restrictions:** You can access geo-blocked content by using a proxy located in a different country.
- **Security (Caching):** Many proxies cache web pages; if multiple users request the same site, the proxy serves the saved copy, saving bandwidth and increasing speed.
- **Content Filtering:** Organizations use them to block access to specific websites for employees or students.

Common Use Cases

- **Web Scraping:** Gathering data from websites without being blocked for too many requests from one IP.
- **Bypassing Censorship:** Accessing information in regions with strict internet firewalls.
- **Security Layering:** Hiding the internal network structure from the public internet.

Disadvantages

- **Speed:** Free or overloaded proxies can significantly slow down your connection.
- **Security Risks:** Since the proxy handles your data, a malicious proxy provider could potentially log, track, or even modify your traffic.
- **Lack of Encryption:** Unlike a VPN, many basic web proxies do not encrypt your traffic between your browser and the proxy server.
- **Compatibility:** Some proxies only work with web traffic (HTTP/HTTPS) and cannot handle other protocols.

What Type of Privacy Does it Provide?

It is important to understand that a proxy is not a privacy silver bullet.

- **IP Masking:** Yes, it hides your public IP address from the destination website.
- **No Full Privacy:** Your ISP can still see that you are connecting to a proxy server (unless you are also using a VPN or HTTPS).
- **Data Exposure:** Because many proxies don't offer end-to-end encryption, your data may be exposed to the proxy operator or anyone sniffing the connection between you and the proxy.
- **Not for Sensitive Tasks:** Avoid using standard proxies for banking, logging into private accounts, or anything involving sensitive credentials, as the proxy provider is essentially an "in-the-middle" party.

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